PARISHRAM 2.0

CLASS 12TH 2026

Inverse Trigonometric Functions

MATHS | LECTURE-3

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TOPICS TO BE COVERED

1. Property Bunch 3 ✓
2. Property Bunch 4 ✓
3. More Questions on Properties ✓

Property 1 and 2

Properties Set 1

(i) $\sin^{-1}(\sin\theta) = \theta$, for all $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

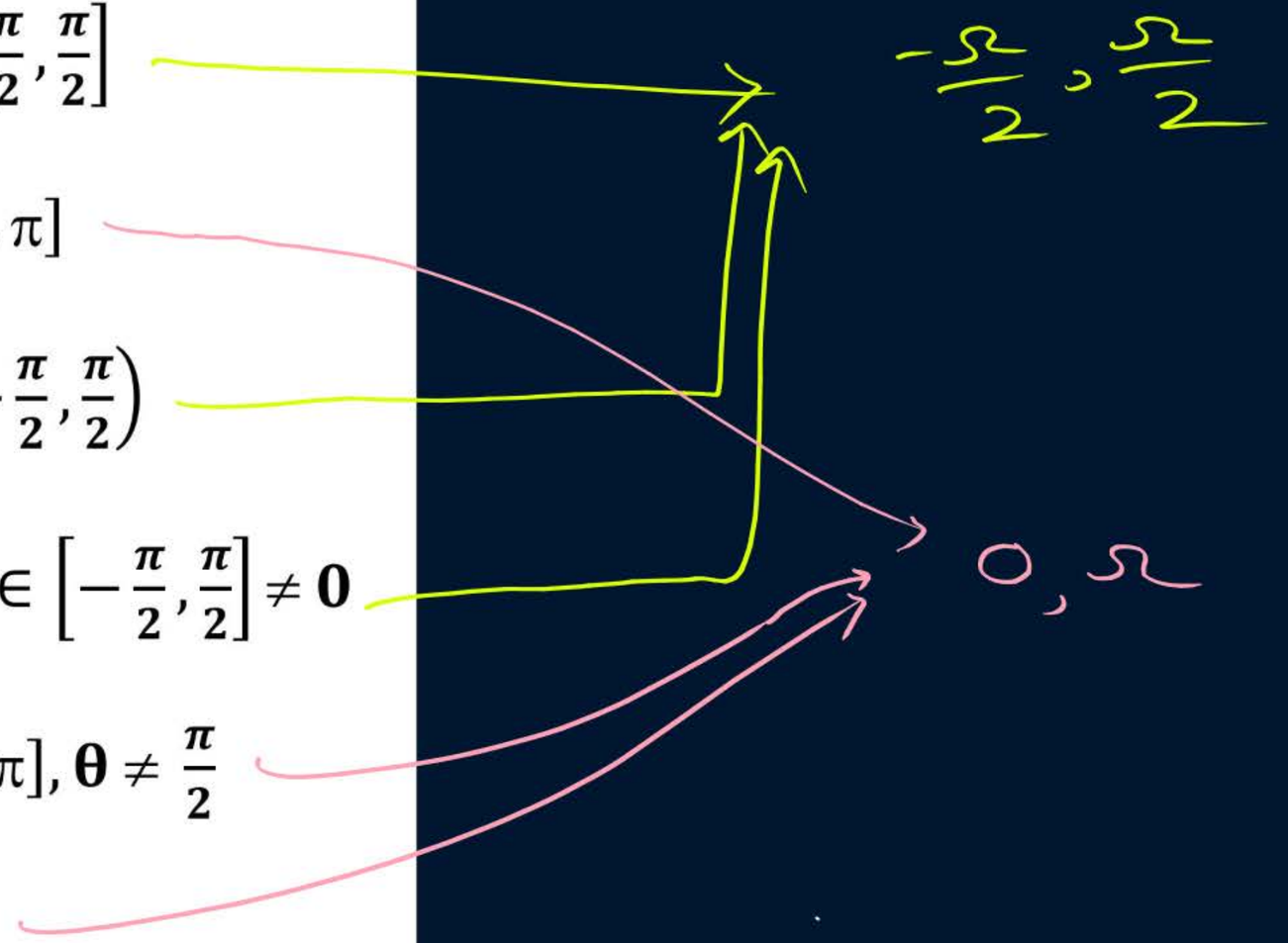
(ii) $\cos^{-1}(\cos\theta) = \theta$, for all $\theta \in [0, \pi]$

(iii) $\tan^{-1}(\tan\theta) = \theta$, for all $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(iv) $\text{cosec}^{-1}(\text{cosec}\,\theta) = \theta$, for all $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \neq 0$

(v) $\sec^{-1}(\sec\theta) = \theta$, for all$\theta \in [0, \pi], \theta \neq \frac{\pi}{2}$

(vi) $\cot^{-1}(\cot\theta)$, for all $\theta \in (0, \pi)$



Properties Set 2

(i) $\sin(\sin^{-1}x) = x$, for all $x \in [-1, 1]$

(ii) $\cos(\cos^{-1}x) = x$, for all $x \in [-1, 1]$

(iii) $\tan(\tan^{-1}x) = x$, for all $x \in R$

(iv) $\text{cosec}(\text{cosec}^{-1}x) = x$, for all $x \in (-\infty, -1] \cup [1, \infty)$

(v) $\sec(\sec^{-1}x) = x$, for all $x \in (-\infty, -1] \cup [1, \infty)$

(vi) $\cot(\cot^{-1}x) = x$, for all $x \in R$

$[-1, 1]$

$x \in R$

Property Bunch 3

(i) $\sin^{-1}(-x) = -\sin^{-1}x,\ x \in [-1, 1]$

(ii) $\cos^{-1}(-x) = \pi - \cos^{-1}x,\ x \in [-1, 1]$

(iii) $\tan^{-1}(-x) = -\tan^{-1}x,\ x \in \mathbf{R}$

(iv) $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x,\ |x| \geq 1$

(v) $\sec^{-1}(-x) = \pi - \sec^{-1}x,\ |x| \geq 1$

(vi) $\cot^{-1}(-x) = \pi - \cot^{-1}x,\ x \in \mathbf{R}$

$\sin^{-1}(-x) = -\sin^{-1}x$

$\tan^{-1}(-x) = -\tan^{-1}x$

$\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x$

$\cos^{-1}(-x) = \pi - \cos^{-1}x$

$\cot^{-1}(-x) = \pi - \cot^{-1}x$

$\sec^{-1}(-x) = \pi - \sec^{-1}x$

Property Bunch 4

(i) $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$, for all $x \in [-1, 1]$

$\sin^{-1}\left(\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$

(ii) $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$, for all $x \in \mathbf{R}$

$\tan^{-1}\left(\sqrt{3}\right) + \cot^{-1}\left(\sqrt{3}\right) = \frac{\pi}{2}$

(iii) $\sec^{-1}x + \operatorname{cosec}^{-1}x = \frac{\pi}{2}$, for all $x \in (-\infty, -1] \cup [1, \infty)$

$\sec^{-1}2 + \operatorname{cosec}^{-1}2 = \frac{\pi}{2}$

POLL QUESTION

If $\sin^{-1}x = y$, then

A $0 \le y \le \pi$

B $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$

C $0 < y < \pi$

D $-\frac{\pi}{2} < y < \frac{\pi}{2}$

$\sin^{-1}x = y$

$y \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$\rightarrow \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

Notes → Live class

Screen

Q → Maths

Solve

Focus [Live class]

Book (Notes)

Notes
↳ Pencil

Stop spamming

After class → QT

POLL QUESTION

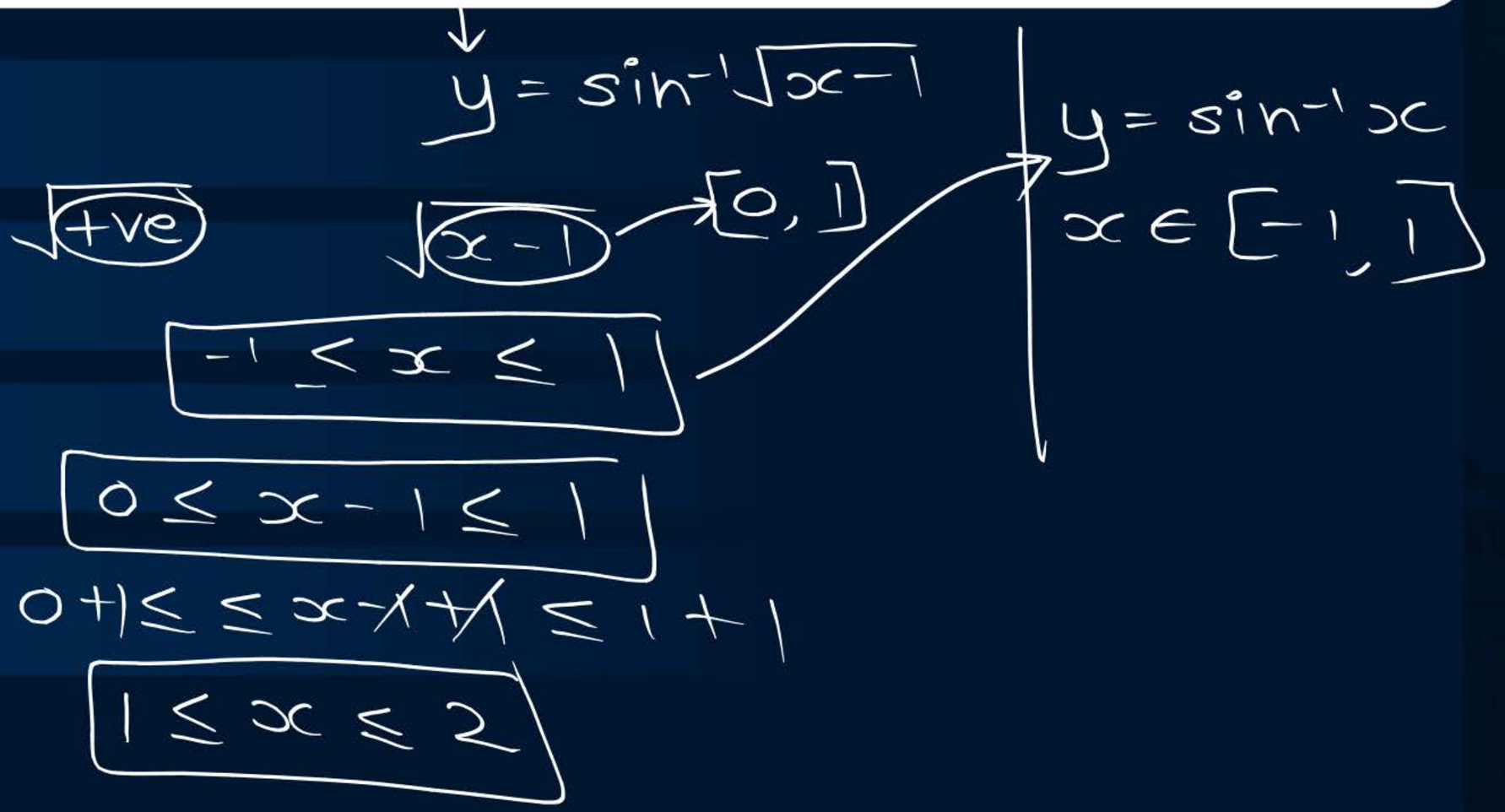
The domain of the function defined by $f(x) = \sin^{-1}\sqrt{x-1}$.

A [1, 2]

B [–1, 1]

C [0, 1]

D None of these



POLL QUESTION

$\cos^{-1}(-x) = \pi - \cos^{-1} x$

The principal value of $\cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)$ is equal to

A $\frac{7\pi}{4}$

B $\frac{-\pi}{4}$

C $\frac{\pi}{4}$

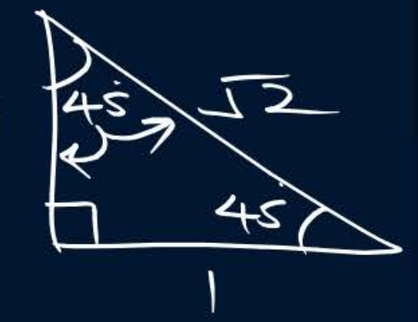
D $\frac{3\pi}{4}$

$\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \pi - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$

$= \pi - \cos^{-1}\left(\cos\frac{\pi}{4}\right)$

$= \pi - \frac{\pi}{4}$

$= \frac{4\pi - \pi}{4} = \frac{3\pi}{4}$



$\cos 45 = \frac{1}{\sqrt{2}}$

$\cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

POLL QUESTION

$\sec^{-1}(-x) = \pi - \sec^{-1}x$

The principal value of $\sec^{-1}\left(\frac{-2}{\sqrt{3}}\right)$ is equal to

A $\frac{5\pi}{6}$

B $\frac{\pi}{6}$

C $\frac{-\pi}{6}$

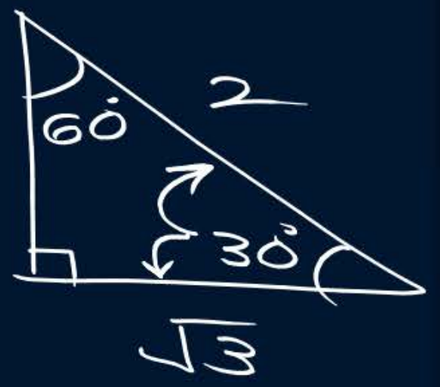
D $\frac{-5\pi}{6}$

$\sec^{-1}\left(\frac{-2}{\sqrt{3}}\right) = \pi - \sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$

$= \pi - \sec^{-1}\left(\sec\frac{\pi}{6}\right)$

$= \pi - \frac{\pi}{6}$

$= \frac{6\pi - \pi}{6} = \frac{5\pi}{6}$



$\sec = \frac{H}{A}$

$\sec 30° = \frac{2}{\sqrt{3}}$

$\sec\frac{\pi}{6} = \frac{2}{\sqrt{3}}$

POLL QUESTION

The principal value of $\tan^{-1}\sqrt{3} - \cot^{-1}(-\sqrt{3})$ is

A $-\pi$

B $-\pi/2$

C $\pi/3$

D $\pi/2$

$$\tan^{-1}(\sqrt{3}) - [\pi - \cot^{-1}(\sqrt{3})]$$

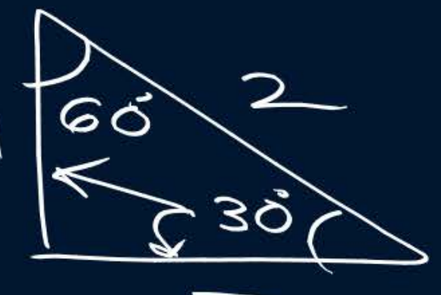
$$\tan^{-1}\left(\tan\frac{\pi}{3}\right) - \left[\pi - \cot^{-1}\left(\cot\frac{\pi}{6}\right)\right]$$

$$= \frac{\pi}{3} - \left[\pi - \frac{\pi}{6}\right]$$

$$= \frac{\pi}{3} - \pi + \frac{\pi}{6}$$

$$= \frac{2\pi}{3\times 2} - \frac{6\pi}{6} + \frac{\pi}{6} = \frac{2\pi}{6} - \frac{6\pi}{6} + \frac{\pi}{6} = \frac{2\pi - 6\pi + \pi}{6}$$

$$= \frac{-3\pi}{6} = -\frac{\pi}{2}$$



$$\tan\frac{\pi}{3} = \tan 60^\circ = \sqrt{3}$$

$$\cot\frac{\pi}{6} = \cot 30^\circ = \sqrt{3}$$

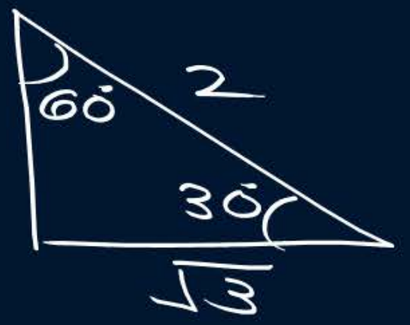
$\cos^{-1}(-x) = \pi - \cos^{-1}x$

QUESTION

Write the principal value of $\tan^{-1}1 + \cos^{-1}\left(-\frac{1}{2}\right)$.

$$\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right)$$

$$= \tan^{-1}\left(\tan\frac{\pi}{4}\right) + \pi - \cos^{-1}\left(\frac{1}{2}\right)$$



$$= \frac{\pi}{4} + \pi - \cos^{-1}\left(\cos\frac{\pi}{3}\right)$$

$$= \frac{\pi}{4} + \pi - \frac{\pi}{3}$$

$$= \frac{\pi}{4} + \frac{3\pi - \pi}{3} = \frac{\pi}{4} + \frac{2\pi}{3} = \frac{3\pi + 8\pi}{12} = \frac{11\pi}{12}$$

$\cos 60° = \frac{1}{2}$

$\cos\frac{\pi}{3} = \frac{1}{2}$

QUESTION

Find the principal value of $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right)$

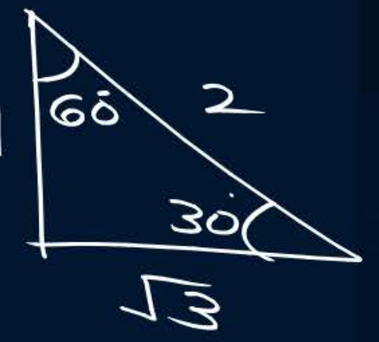
$$\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) = \pi - \cot^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= \pi - \cot^{-1}\left(\cot\frac{\pi}{3}\right)$$

$$= \pi - \frac{\pi}{3}$$

$$= \frac{3\pi - \pi}{3}$$

$$= \frac{2\pi}{3}$$



$\cot 60° = \frac{1}{\sqrt{3}}$

$\cot\frac{\pi}{3} = \frac{1}{\sqrt{3}}$

QUESTION

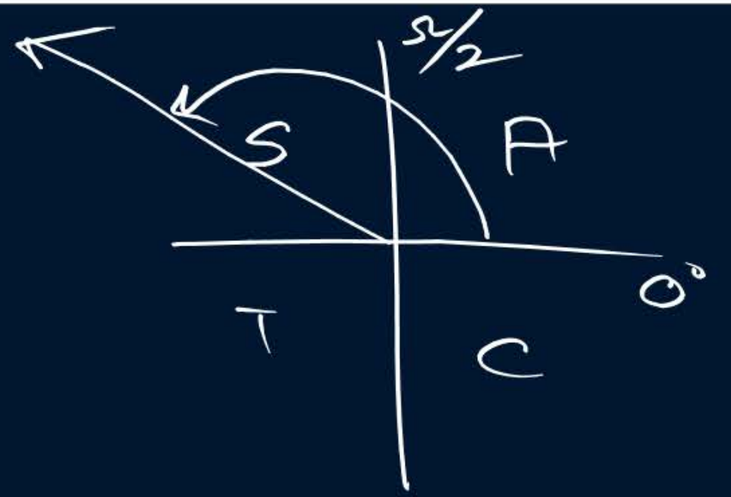
$\frac{\pi}{2} \rightarrow$ Trigo change

Simplify : $\cos\left(\frac{\pi}{2} + \sin^{-1}\frac{1}{\sqrt{3}}\right)$ Show your work.

$$\cos\left(\frac{\pi}{2} + \sin^{-1}\frac{1}{\sqrt{3}}\right)$$

$$= -\sin\left(\sin^{-1}\frac{1}{\sqrt{3}}\right)$$

$$= -\frac{1}{\sqrt{3}}$$



THANK YOU